

# Lavoisier: a critical review

Evidence, assurance boundaries and the next design decision

Project review

8 October 2026

# The decision in one slide

**Keep the research result. Harden the assurance boundary. Test a simpler hybrid.**

- Strong prototype for small, explicitly modelled resource flows.
- Useful compiler checks; incomplete evidence for general engineering assurance.
- The biggest next gains are trustworthy accounting, conformance and usability.
- Compare alternatives on the same problem before expanding or rewriting.

Reviewed revision: 43d1f78014f52e01ae8bbf60fe234d821209a600. Full evidence: CRITICAL\_REVIEW\_REPORT.md.

# What is strong

- **Evidence trail:** experiments, negative results, findings and rule changes.
- **Useful static rejection:** wrong states, duplicate use of one token, unit mismatches.
- **Meaningful tests:** success, retry and failure accounting; negative examples.
- **Inspectable outputs:** ordinary Rust, documented assumptions, generated views.

Report §3; E1–E5, E9–E13. These strengths justify continuing the project.

# The objective determines the design

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Original experiment

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How far can Rust's types go?  
Prefer more compiler checking

Record where machinery breaks

Type complexity can be evidence

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An engineering product

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Which decisions become more reliable?  
Minimise total modelling and review effort

Support real inputs, change and uncertainty

Type complexity needs a measurable payoff

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**The present design fits the first objective better than the second.**

Report §§1, 5; instructions.md, Summary and R8/R13. A change in objective should be explicit.

# Fresh validation: a green baseline

| Review run            | Result                                              |
|-----------------------|-----------------------------------------------------|
| Model CI              | All eight gates passed                              |
| Model test functions  | 182 passed; 0 failed; 0 ignored                     |
| Nested trybuild cases | 50 expected rejections matched                      |
| Doctests              | 64 passed; includes expected failures/panics        |
| Tooling tests         | 31 spec-gen + 4 diagram-gen passed                  |
| Specs / analysis      | 6 clean specs; 0 errors, 4 warnings, 125 info items |

Linux, Rust 1.98.1 with rust-src and Clippy. Nested trybuild cases are not additional top-level test functions. Report §2.

# A green gate has several meanings

| Layer                   | What the evidence actually supports            |
|-------------------------|------------------------------------------------|
| Rust types              | Encoded state, unit and ownership constraints  |
| Full compilation        | Written const assertions for instantiated code |
| Executed tests          | Observed behaviours and selected paths         |
| Source tags and scanner | Declared links and recognised source forms     |
| Measurements            | Physical validity within a calibrated range    |

**Passing one layer does not establish the next.**

Report §§2, 4. Physical validation remains open; source tags do not establish test execution.

# Reproduced gap: an executed loss can pass

## **A public supplier can expose an item with no tripwire.**

- ① Draw one DryTeabag from a one-item box.
  - ② Move the bag into a named underscore binding.
  - ③ Let it leave scope without returning it or using a consumer.
- Test passes; project Clippy restrictions also pass.
  - No unsafe code, explicit drop, forget or test-support feature.
  - This extends beyond the familiar untested-branch and panic gaps.

R01 / P1; resource.rs, tea resources. Isolated public-API probe; not a claim that the shipped tea flow loses a bag.

# Reproduced gaps: evidence needs a trust boundary

| Probe                                | Observed result                                         | Meaning                         |
|--------------------------------------|---------------------------------------------------------|---------------------------------|
| Ignored test with Verifies:          | Trace script says verified; runner: 0 passed, 1 ignored | A tag is a declared link        |
| Downstream Recordable implementation | Invented labour event enters History                    | Event truth trusts implementors |

## Track executed evidence and trusted event producers explicitly.

- Baseline tests were not ignored.
- These probes test mechanisms, not the full gate for a new model crate.

R03/P2 and R08/P4; trace.sh and history.rs. Reproduction details: report §9.



# Reproduced workflow defect: scaffold overwrite

**A “one-shot” generator accepts an existing output directory.**

- ① Generate a CS-1 scaffold in a disposable directory.
- ② Add a comment representing a hand-maintained edit.
- ③ Rerun the identical command.

**Observed:** exit 0; the hand edit disappears.

**Fix:** refuse existing output by default; preflight and stage writes; make replacement explicit.

R04/P3; specgen.rs output loop. Probe stayed outside the repository. No existing project file was overwritten.

# Coverage and conformance are the next assurance gaps

- **Fixed analysis list:** a new workspace member does not automatically join the lint scope.
- **Separate truths:** a clean spec and clean Rust model can still describe different quantities.
- **Trusted model code:** crate-visible mint/defuse and handwritten assertions define the real boundary.
- **Shared parser:** matching diagrams and reports may share the same extraction error.

**Next:** discover membership, measure extraction coverage, and compare stable semantic contracts.

R02, R05–R07. Source-inspected architectural risks; no current model-crate omission claimed.

# Where the modelling promise ends

| Useful property         | Further evidence required       |
|-------------------------|---------------------------------|
| Balanced energy         | Validated thermal model         |
| One token used once     | Unique real-resource identity   |
| Labour budget conserved | Feasible calendar and deadlines |
| Same final types        | Behavioural / temporal analysis |
| Compatible units        | Meaningful physical operations  |

**Consistency does not establish physical validity.**

R09–R10. Temperature is not an additive conserved amount; a boiling state is a declared model state.

# The cost is not just compilation time

- Unary inventory and recursive lists have a documented scale ceiling.
- Callers maintain exact balances; failure paths multiply types and signatures.
- Source conventions and custom parsing create continuing maintenance work.
- New users must learn Rust and the project's exceptions to normal idioms.

**Scaffold survival is not effort saved:** 697 / 1,154 generated lines survived (60.4%); that is about 27.4% of the final model's lines.

R11; F-010–012, F-030, F-034, F-066. Historical scale measurements were not rerun in this review.

# Options: retain, simplify, or formalise the contract

| Option                            | Strongest benefit                             | Principal cost                      |
|-----------------------------------|-----------------------------------------------|-------------------------------------|
| A. Harden current Rust            | Preserve static checks and evidence           | Existing scale/ergonomic limits     |
| B. Typestate + ledger             | Dynamic quantities, explicit final accounting | More runtime checks; trusted ledger |
| C. Versioned model representation | Conformance, diagnostics, multiple views      | Schema and language/tool ownership  |

**Recommendation:** do A now; compare a small B prototype; introduce only the portion of C the comparison needs.

Report §6. These are design assessments, not measured benchmark rankings.

# Options: use the tool that answers the question

| Question                            | Candidate approach              | Main limit                  |
|-------------------------------------|---------------------------------|-----------------------------|
| Feasible amounts or schedules?      | SMT / constraint programming    | Encoding and search cost    |
| Deadlock or reachable bad state?    | Petri nets / model checking     | State explosion and bounds  |
| Physical dynamics or domain review? | Simulation / MBSE workflow      | Calibration and integration |
| Stronger universal guarantees?      | Linear/dependent types / proofs | Expertise and migration     |

**No alternative automatically supplies all of these guarantees.**

Options D–G, report §6. TLA+, Alloy, Petri nets and optimisation solve different classes of problem.

# A hybrid worth testing

**Approved contract** → **early checks** → **Rust states + accounting ledger**

- Stable IDs, units, balances, requirements and assumption provenance.
- Checked transactions for resource creation, transfer and final disposition.
- Optional solver, temporal or simulation adapters for specific questions.
- Evidence record: revision, tools, executed checks and unresolved obligations.
- Generated views expose assumptions and extraction coverage.

Report §7. A ledger adds obligations: identity, atomicity, rollback, bypass prevention and independent validation.

# What to improve first

| Change, in priority order   | Acceptance evidence               |
|-----------------------------|-----------------------------------|
| Account for extracted items | Permanent P1 regression/contract  |
| Protect scaffold output     | Refuse existing files             |
| Track executed verification | Ignored tests do not qualify      |
| Discover model membership   | New models cannot escape analysis |
| Broaden reproducible gate   | Tools, compiler and conformance   |

Report §8. No fixes to existing files were made during this review.



# Make the next experiment comparative

## **Same CS-6 slice. Same obligations. Independent violation cases.**

- Compare current Rust with a small ledger-based version.
- Include unseen errors: loss, overdraw, wrong state, failure accounting and spec drift.
- Measure authoring/change time, diagnostic comprehension and false positives.
- Measure build time, memory and runtime cost as inventory grows.
- Use independent modellers; agree decision thresholds before running.

Report §8. Do not compare against a weaker specification or use lines of code as the primary success measure.

# Put one real process under measurement

- Separate conserved amounts, state variables, capacities, costs and observations.
- Give constants a source, uncertainty, calibration date and validity range.
- Agree acceptance tolerances before gathering validation data.
- Use held-out observations to test the physical model.

**A passed conservation check is one input to engineering confidence.**

R09; report §§4 and 8. Physical validation is a proposed next step, not a completed review result.

# Choose a bounded future

**Near term:** a dependable specialist resource-flow checker and research corpus.

**Expand when:** independent users and comparative results demonstrate value.

**Integrate when:** scheduling, physical simulation or enterprise evidence workflows dominate the need.

**Defer:** more similar case studies, a whole-model macro language, or a host-language rewrite without a discriminating experiment.

Report §§8 and 11. Keep the current implementation as the control and preserve its negative findings.

# Review basis and limits

- Fresh baseline: eight CI gates, 182 test functions, 64 doctests and 35 tooling tests passed.
- Four disposable probes: resource loss, ignored verification, output overwrite and trusted history extension.
- Source inspection: kernel, representative models, gates, scanner and generator.
- Not performed: exhaustive audit, new scale benchmarks, physical calibration or alternative-tool benchmarks.
- External reference sites were blocked; option descriptions are not a current tool-compatibility survey.

Report §§2, 9–10 contain commands, scope limits, evidence links and further reading.

# Recommended next decision

**Continue, with a precise promise and a measured comparison.**

- ① Harden accounting, evidence reporting and generation safety.
- ② Compare a small hybrid on the same specification.
- ③ Validate one physical process with independent evidence.
- ④ Select the product boundary from those results.

Full report: [CRITICAL\\_REVIEW\\_REPORT.md](#).

The next success metric should be better engineering decisions per unit of modelling effort.